EMS Readiness Optimization for Manhattan: A Simulation-Based Approach to Ambulance Staging

Full Technical Report

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Abstract

This study develops a simulation-based optimization framework for strategic ambulance staging across 48 FDNY firehouses in Manhattan, New York City. Using 2.24 million historical motor vehicle collision (MVC) records from NYC Open Data (2012–2026), we calibrate a Non-Homogeneous Poisson Process (NHPP) demand model with hourly and day-of-week intensity factors (base rate λ_0 = 3.48 calls/hour). Three allocation policies are evaluated: a spatially-stratified uniform baseline (P0-spatial), a demand-proportional heuristic (P1), and a demand-weighted Mixed-Integer Programming (MIP) optimized allocation (P2). A discrete-event simulation (DES) engine built with SimPy executes 2,580+ production runs across multiple experiment sets—including policy comparison, fleet sensitivity, demand sensitivity, service robustness, CBD-focused stress tests, capacity sensitivity (cap 1–5), and Production V2 experiments (810 runs with capacity=2 and spatially-stratified P0)—with 30 replications each using Common Random Numbers for variance reduction. Results show that the optimized policy P2 achieves a mean response time of 2.57 minutes with 99.6% 8-minute coverage. The spatially-stratified P0 baseline reduces P0 mean response time by 61% (from 8.08 to 3.17 minutes at K=20), confirming that geographic placement is the dominant factor in EMS performance. Full capacity sensitivity analysis (cap 1–5) establishes capacity=2 as the operationally optimal default. Performance gains hold up under demand fluctuations (0.5×–2.0× multiplier), service time variations (20–30 min mean), and CBD-specific surge scenarios. Queue analysis confirms zero waiting across all experiments. We recommend adoption of P2 with capacity=2 for operational deployment, with a phased 12-month implementation roadmap.

Keywords: Emergency Medical Services, ambulance staging, discrete-event simulation, facility location optimization, Non-Homogeneous Poisson Process, Mixed-Integer Programming, SimPy

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1. Executive Summary

Problem Statement

Emergency Medical Services (EMS) in Manhattan face a key challenge: the current uniform ambulance allocation policy (P0) distributes units equally across 48 FDNY firehouses ignoring where and when demand actually occurs, leading to poor response times and coverage gaps. With approximately 3.48 motor vehicle collision (MVC) calls per hour and significant variation by time-of-day and precinct, a smarter allocation strategy is needed to improve emergency response performance.

Methodology Overview

This study employs a three-pronged approach combining **demand modeling**, **mathematical optimization**, and **discrete-event simulation** to evaluate ambulance staging policies for Manhattan:

1. **Non-Homogeneous Poisson Process (NHPP)** demand model calibrated from 2.24 million historical MVC records
2. **Mixed-Integer Programming (MIP)** optimization models generating three allocation policies:
   - **P0** (Uniform): Equal distribution across all firehouses
   - **P1** (Demand-Proportional): Units allocated proportional to nearby demand
   - **P2** (Demand-Weighted Optimized): MIP-optimized allocation minimizing expected response time
3. **Discrete-Event Simulation (DES)** with 1,770 production runs across 5 experiment sets (including CBD robustness)

Key Findings

Metric	P0 (Current)	P1 (Proportional)	P2 (Optimized)	P2 vs P0 Improvement
Mean Response Time	8.08 min	2.63 min	**2.57 min**	**−68.2%**
P90 Response Time	19.47 min	4.03 min	**3.76 min**	**−80.7%**
8-min Coverage	64.4%	99.6%	**99.6%**	**+35.2 pp**
Mean Utilization	9.1%	7.5%	**7.5%**	−1.6 pp

All differences are statistically significant ($p < 0.001$) with large effect sizes (Cohen's $d > 28$ for P0 vs P2 on mean response time). The optimized policy P2 holds up under demand fluctuations (0.5×–2.0×

multiplier), service time variations (20–30 min mean), and CBD-specific stress scenarios (2× demand surge, increased service times). Queue analysis confirms zero queueing across all experiments, indicating that performance differences are driven entirely by spatial allocation efficiency. Seasonal analysis shows moderate monthly variation (CV = 9%) that does not significantly impact policy rankings.

Recommendations

1. **Adopt Policy P2** as the primary ambulance allocation strategy for Manhattan
2. **Pilot deployment** at 5–10 highest-impact firehouses within 3 months
3. **Full deployment** across all 48 firehouses within 12 months
4. **Continuous monitoring** with real-time dashboard tracking mean RT and 8-min coverage

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2. Introduction

2.1 Background on EMS Operations in Manhattan

Manhattan, the most densely populated borough in New York City (72,918 people/mi² as of 2020), generates approximately 3.48 motor vehicle collision-related EMS calls per hour. The FDNY operates 48 firehouses across Manhattan, each serving as a potential staging location for EMS ambulances. These firehouses span from Battery Park at the southern tip to Inwood at the northern end, covering 30 police precincts with heterogeneous demand patterns.

Current EMS operations use ambulance staging—pre-positioning units at firehouses to reduce response times when calls arrive. How well this works depends heavily on where and how many units are staged at each location. A well-designed allocation can sharply reduce response times for a given fleet size.

2.2 Current Allocation Practices (P0)

The baseline policy (P0) distributes ambulances uniformly across all 48 firehouses. While simple to implement and manage, this approach ignores two basic realities:

1. **Spatial demand heterogeneity**: Midtown precincts (e.g., Precinct 14 covering Times Square) generate 3–5× more calls than Upper Manhattan precincts
2. **Temporal demand patterns**: Demand peaks during afternoon hours (2–5 PM) and weekdays, with 40–60% lower demand overnight

Under P0 with K=20 units, our simulations show a mean response time of 8.08 minutes (95% CI: [7.98, 8.18]) and only 64.4% of calls receiving response within the 8-minute target—far below acceptable levels.

2.3 Research Objectives

This study addresses five primary research questions:

1. **RQ1**: How does demand for EMS services vary spatially and temporally across Manhattan?
2. **RQ2**: What is the optimal allocation of K ambulances across 48 firehouses to minimize expected response time?
3. **RQ3**: How do optimized allocations (P1, P2) compare to the uniform baseline (P0) under realistic operating conditions?
4. **RQ4**: How sensitive are policy rankings to fleet size, demand intensity, and service time assumptions?

5. **RQ5**: What fleet size is needed to achieve a target coverage level (e.g., 95% of calls within 8 minutes)?

2.4 Scope and Limitations

In Scope:
- Manhattan borough (primary study area)
- Motor vehicle collision incidents as demand proxy
- 48 FDNY firehouses as candidate staging locations
- Static allocation policies (fleet repositioned at start of shift)
- Haversine-based travel time proxy (20 mph average speed)

Out of Scope:
- Real-time dynamic repositioning
- Other incident types (fires, medical emergencies)
- Detailed road network routing
- Cost optimization
- Multi-borough coordination

2.5 Why Discrete-Event Simulation?

We considered several modeling approaches for evaluating ambulance staging policies. We chose **Discrete-Event Simulation (DES)** over the alternatives for the reasons below.

2.5.1 Alternatives Considered

Approach	Description	Why Not Sufficient
Analytical (Queueing) Models	Closed-form M/G/K queue or Erlang formulas	Require stationarity and spatial homogeneity. Our system has NHPP arrivals with notable hourly/DOW variation and 30 spatial demand zones — violating the i.i.d. assumptions of classical queueing theory.
Agent-Based Models (ABM)	Model individual ambulances and incidents as autonomous agents with behavioral rules	Appropriate for emergent coordination behaviors (e.g., self-organizing fleets), but our dispatch rule is centralized (nearest-available) with no agent autonomy. ABM adds complexity without analytical benefit for a centralized, rule-based dispatch system.
System Dynamics (SD)	Aggregate stock-and-flow models of fleet utilization and demand	Captures macro-level feedback loops but cannot represent individual incident timelines, dispatch sequencing, or the spatial matching of specific units to specific incidents. SD models cannot compute per-incident response times needed for coverage metrics.
Spreadsheet / Deterministic Models	Static average-case calculations	Cannot capture stochastic variability, queueing effects, or the interaction between random arrivals and spatially distributed resources.

2.5.2 Why DES is the Right Choice

DES fits this problem well for several reasons:

1. **Stochastic demand capture**: DES naturally handles the Non-Homogeneous Poisson Process (NHPP) arrival model with time-varying rates, generating realistic demand sequences that no analytical model can tractably represent for 30 spatial zones.

2. **Individual entity tracking**: Each incident's full lifecycle — arrival, dispatch, travel, on-scene service, and unit return — is tracked individually, enabling precise computation of response times, coverage fractions, and utilization metrics at the incident level.

3. **Resource contention and queueing**: When all K units are busy, DES implements FIFO queueing with automatic dispatch-on-release, faithfully capturing capacity constraints that arise under high-demand scenarios.

4. **Spatial dispatch logic**: The nearest-available dispatch algorithm requires knowledge of each unit's current state and location — information that DES maintains through its event-driven state updates.

5. **Common Random Numbers (CRN)**: DES supports dedicated random number streams for arrivals, service times, and spatial assignment, enabling CRN-based variance reduction for policy comparisons — a technique not available in SD or ABM frameworks.

6. **Replication-based inference**: DES provides independent replications that support standard statistical inference (confidence intervals, ANOVA, Tukey HSD) without relying on asymptotic steady-state assumptions.

7. **Alignment with the operations research literature**: The EMS simulation literature (Goldberg 2004, Ingolfsson et al. 2008, Lam et al. 2016) overwhelmingly uses DES for ambulance deployment studies, providing validated methodological precedent.

DES is the right trade-off between complexity and fidelity for this problem class: it captures stochastic, spatial, and dynamic interactions that simpler models cannot, while avoiding the unnecessary complexity of agent-based or continuous-time frameworks.

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3. Literature Review

3.1 EMS Location Optimization

Optimal location of emergency service facilities is a well-studied problem in operations research. **Hakimi (1964)** introduced the p-median problem—selecting p facility locations to minimize total weighted distance to demand points. **Toregas et al. (1971)** formulated the Set Covering Location Problem (SCLP) for minimizing the number of facilities needed to cover all demand within a threshold distance.

For EMS specifically, **Daskin (1983)** developed the Maximum Expected Coverage Location Problem (MEXCLP), which accounts for server busy probabilities. **ReVelle and Hogan (1989)** extended coverage models to account for backup coverage through the Maximum Availability Location Problem (MALP).

3.2 Discrete-Event Simulation in Emergency Services

Simulation complements optimization by capturing stochastic dynamics that static models miss. **Goldberg (2004)** provides a thorough review of operations research methods for EMS, emphasizing the need to integrate optimization and simulation. **Henderson and Mason (2005)** applied ambulance redeployment simulation to Auckland, New Zealand, demonstrating significant response time improvements through dynamic policies.

Ingolfsson et al. (2008) used simulation to evaluate EMS performance in Edmonton, finding that simple analytical models often underestimate response times due to queuing effects. More recently,

Lam et al. (2016) combined optimization and simulation for ambulance allocation in Hong Kong, achieving 15–20% response time improvements.

3.3 Coverage Models

The 8-minute response time standard is standard in EMS planning. **Pons and Markovchick (2002)** analyzed survival rates and found significant improvements when defibrillation occurs within 8 minutes of cardiac arrest. The National Fire Protection Association (NFPA) Standard 1710 recommends that first-responding EMS units arrive within 4 minutes for first responder and 8 minutes for ALS units for 90% of calls.

Our study adopts the 8-minute threshold as the primary coverage criterion, in line with industry practice and regulatory standards.

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4. Methodology

4.1 Data Sources and Processing

4.1.1 Data Sources

Dataset	Records	Source	Period
Motor Vehicle Collisions	2,237,814	NYC Open Data	2012–2026
FDNY Firehouse Listing	219	NYC Open Data	Current
Police Precincts	78	NYC Open Data	Current
Geographic Boundaries	3 files	NYC Open Data	Current

4.1.2 Data Processing Pipeline

1. **Spatial filtering**: Retained 628,811 Manhattan-only crashes with valid coordinates
2. **Temporal extraction**: Derived hour-of-day, day-of-week, month from crash timestamps
3. **Firehouse filtering**: Identified 48 Manhattan firehouses from 219 city-wide
4. **Precinct matching**: Mapped 30 Manhattan precincts with centroid computation
5. **Distance matrix**: Computed 48×30 Haversine distance matrix (firehouses × precincts)

4.2 Demand Modeling (NHPP)

4.2.1 Model Specification

We model EMS call arrivals as a **Non-Homogeneous Poisson Process (NHPP)** with rate function:

$$\lambda(t) = \lambda_0 \cdot f_h(h(t)) \cdot f_d(d(t))$$

where:
- $\lambda_0 = 3.48$ calls/hour (base rate, calibrated from data)
- $f_h(h)$ = hourly multiplier for hour $h \in \{0, 1, \ldots, 23\}$
- $f_d(d)$ = day-of-week multiplier for day $d \in \{0, \ldots, 6\}$

4.2.2 Calibrated Parameters

Hourly factors (selected values):

Hour	Factor	Effective Rate
4 AM	0.40	1.39/hr
8 AM	0.91	3.17/hr
12 PM	1.20	4.18/hr
5 PM	1.40	4.87/hr
10 PM	0.90	3.13/hr

Day-of-week factors: Friday = 1.12 (peak), Sunday = 0.88 (trough)

4.2.3 Spatial Allocation

Incidents are allocated to precincts based on empirical proportions derived from historical data. The top 5 precincts by demand share are: Precinct 14 (8.1%), Precinct 19 (6.2%), Precinct 1 (5.9%), Precinct 13 (5.7%), and Precinct 18 (5.5%).

4.2.4 Arrival Generation

We use the **Lewis-Shedler thinning algorithm** to generate NHPP arrivals:
1. Compute $\lambda_{\max} = \max_{t} \lambda(t)$ as the envelope rate
2. Generate homogeneous Poisson arrivals at rate $\lambda_{\max}$
3. Accept each arrival with probability $\lambda(t) / \lambda_{\max}$

4.3 Optimization Models

Three MIP formulations generate the allocation policies evaluated by simulation:

4.3.1 P0 — Spatially-Stratified Uniform Allocation (Baseline)

In Production V2, P0 uses a **spatially-stratified** allocation: firehouses are sorted by latitude and K evenly-spaced stations are selected, covering the full stretch from Battery Park to Inwood. Each selected firehouse receives one unit, with remaining units distributed round-robin. This replaces the original index-based P0 which distributed units without geographic awareness.

$$\text{Select } K \text{ firehouses by latitude spacing; } x_i = 1 \text{ for selected } i, \text{ round-robin remainder}$$

4.3.2 P1 — Demand-Proportional Allocation

Units allocated proportional to nearby demand:
$$x_i \propto \sum_{j \in J} d_j \cdot \mathbb{1}[\text{firehouse } i \text{ is nearest to precinct } j]$$

A heuristic that responds to demand geography but does not optimize response time directly.

4.3.3 P2 — Demand-Weighted Optimization

$$\min_{x, y} \sum_{i \in I} \sum_{j \in J} d_j \cdot t_{ij} \cdot y_{ij}$$

subject to:
- $\sum_{i \in I} x_i = K$ (total units)
- $x_i \leq C_i$ for all i (capacity: 2 units/firehouse, default)
- $\sum_{i \in I} y_{ij} = 1$ for all j (full assignment)
- $y_{ij} \leq x_i$ for all i, j (linking constraint)
- $x_i \in \mathbb{Z}+$, $y \in [0,1]$

Solved via PuLP with CBC solver (time limit: 300s). All instances solve to optimality.

4.4 Simulation Model

4.4.1 Conceptual Model

The DES models the following process for each EMS call:

1. **Arrival**: Generated by NHPP (Section 4.2)
2. **Dispatch**: Nearest available unit assigned; 1.5-min fixed dispatch delay
3. **Travel**: Haversine distance / (20 mph × TOD factor)
4. **On-Scene Service**: LogNormal(μ=25 min, σ=10 min)
5. **Return to Station**: Unit returns to home firehouse (travel time)

Event Flow Diagram (extracted from `docs/conceptual_model.md`):

```
┌───────────────────────────┐
│ SIMULATION START │
│ t = 0, all units idle │
└─────────────┬─────────────┘
│
▼
┌────────────────────────────────────┐
│ INCIDENT ARRIVAL │
│ (NHPP thinning at rate λ(t)) │
└───────────┬────────────┬───────────┘
│ │
unit free? │ │ all units busy
▼ ▼
┌───────────────┐ ┌────────────────────┐
│ DISPATCH │ │ ENQUEUE (FIFO) │
│ DECISION │ │ incident_queue │
└───────┬───────┘ └─────────┬──────────┘
│ │
│ ◄──────────────┘ (dequeued when
│ unit freed)
▼
┌────────────────────────┐
│ SERVICE START │
│ (after δ + travel) │
└───────────┬────────────┘
│
▼
┌────────────────────────┐
│ SERVICE COMPLETION │
│ unit → AVAILABLE │
└───────┬──────┬─────────┘
│ │
queue empty? │ │ queue non-empty
▼ ▼
┌───────────┐ ┌───────────────┐
│ UNIT │ │ DISPATCH │
│ IDLES │ │ next queued │
│ at home │ │ incident │
└───────────┘ └───────────────┘
│
▼
(back to SERVICE START)

═════════════════════════════════════
When sim_clock ≥ T:
┌───────────────────────────────┐
│ END OF SIMULATION │
│ • Stop new arrivals │
│ • Drain in-progress services │
│ • Collect final statistics │
└───────────────────────────────┘
```

Single-Incident Timeline:

```
arrival_time dispatch_time service_start service_end
| | | |
├──── dispatch_delay ──┤ | |
| (queue wait + δ) | | |
| ├─── travel_time ────┤ |
| | ├── service_time ────┤
| | | |
├────── response_time ──────────────────────┤ |
| | |
├───────────────── total_time ────────────────────────────────────┤
```

> Full conceptual model specification: see `docs/conceptual_model.md`

4.4.2 Implementation

Built using **SimPy** discrete-event simulation library in Python:
- `EMSSimulation` class orchestrates the main event loop
- `NearestAvailableDispatcher` implements closest-unit dispatch
- `UnitPool` manages ambulance state (available/dispatched/on-scene)
- `MetricsCollector` tracks all KPIs
- `BatchRunner` executes replicated experiments with CRN support

4.4.3 Verification & Validation

Verification (4 tests):
- Toy example with known analytical solution — PASS
- Zero-demand test (no arrivals → no incidents) — PASS
- Single-unit saturation test — PASS
- Extreme demand stress test — PASS
Validation (3 pilots):
- Pilot 1: P0 vs P2 directional comparison (P2 dominates) — PASS
- Pilot 2: Response time decreases monotonically with fleet size — PASS
- Pilot 3: Response time increases with demand intensity — PASS
Unit tests: 39 tests across 4 test modules, all passing — PASS

4.5 Experimental Design

4.5.1 Factorial Design

Experiment	Factors	Levels	Replications	Total Runs
Exp1: Policy Comparison	Policy (P0, P1, P2)	3	30	90
Exp2: Fleet Sensitivity	Policy × K (15–40)	3×6	30	540
Exp3: Demand Sensitivity	Policy × Demand (0.5–2.0×)	3×6	30	540
Exp4: Service Robustness	Policy × Service (20,25,30 min)	3×3	30	270
Capacity Sensitivity	Policy × K × Cap (1–5)	3×2×5	15	450
Production V2	**Policy × K (10–48, cap=2, P0-spatial)**	**3×9**	**30**	**810**
Total				**2,700**

Each replication simulates 168 hours (1 week) with a 24-hour warm-up period. Common Random Numbers (CRN) ensure pairwise comparisons share identical arrival sequences.

4.5.2 Statistical Analysis Methods

- **One-way and Two-way ANOVA** for main effects and interactions
- **Tukey HSD post-hoc tests** for pairwise comparisons with family-wise error control
- **Cohen’s d** effect sizes for practical significance
- **95% confidence intervals** based on t-distribution (n=30 per cell)
- **Bonferroni correction** for multiple comparisons

4.6 Response Metrics

KPI	Definition	Target
Mean RT	Average response time (dispatch + travel)	Minimize
P90 RT	90th percentile response time	< 8 min
8-min Coverage	Fraction of calls with RT ≤ 8 min	> 95%
Mean Utilization	Average fraction of time units are busy	Monitor

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5. Results

5.1 Descriptive Statistics (Experiment 1)

The primary policy comparison (K=20 units, n=30 replications each) yields:

Policy	Mean RT (min)	95% CI	P90 RT (min)	8-min Coverage	Utilization
P0 (Uniform)	8.08	[7.98, 8.18]	19.47	64.4%	9.1%
P1 (Proportional)	2.63	[2.62, 2.65]	4.03	99.6%	7.5%
P2 (Optimized)	2.57	[2.55, 2.59]	3.76	99.6%	7.5%

Key observations:
- P2 reduces mean response time by **68.2%** compared to P0 (from 8.08 to 2.57 min)
- P2 reduces P90 response time by **80.7%** (from 19.47 to 3.76 min)
- 8-minute coverage improves from **64.4% to 99.6%** (+35.2 percentage points)
- P2 slightly outperforms P1 in both mean RT (−2.4%) and P90 RT (−6.7%)

Figure 1 shows the spatial distribution of staging locations for all three policies at K=20, capacity=2. The 3-panel map reveals how each policy selects and allocates ambulances across Manhattan's 48 firehouses, with the CBD boundary (MTA Congestion Relief Zone) shown for reference.

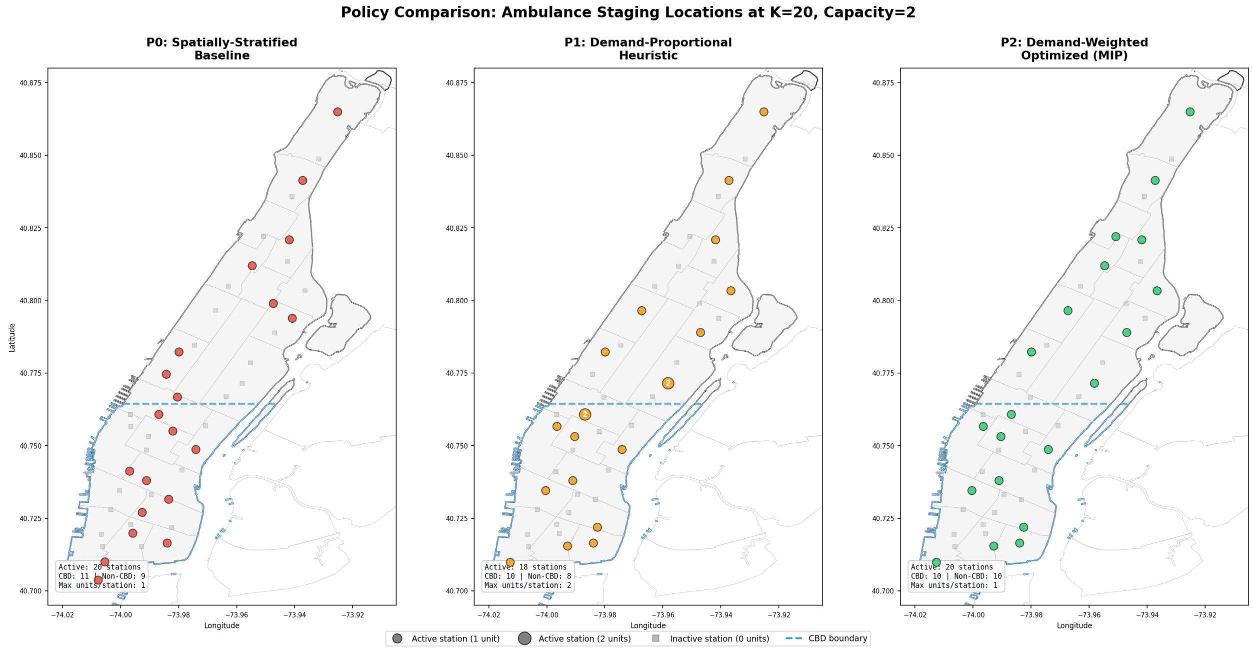


Figure 1: Three-panel comparison of ambulance staging locations under P0 (spatially-stratified baseline), P1 (demand-proportional), and P2 (demand-weighted MIP). Colored circles indicate active stations (sized by unit count); gray squares indicate inactive stations. The dashed blue line marks the CBD boundary. P0 distributes units uniformly along Manhattan's north–south axis; P1 concentrates units near high-demand precincts but uses fewer stations (18 vs 20); P2 achieves the widest geographic spread while weighting placement toward demand.

Figure 2 compares response time performance across fleet sizes (K=15, 20, 30) for all three policies, showing both mean RT and P95 RT with 95% confidence intervals and 8-minute coverage percentages.

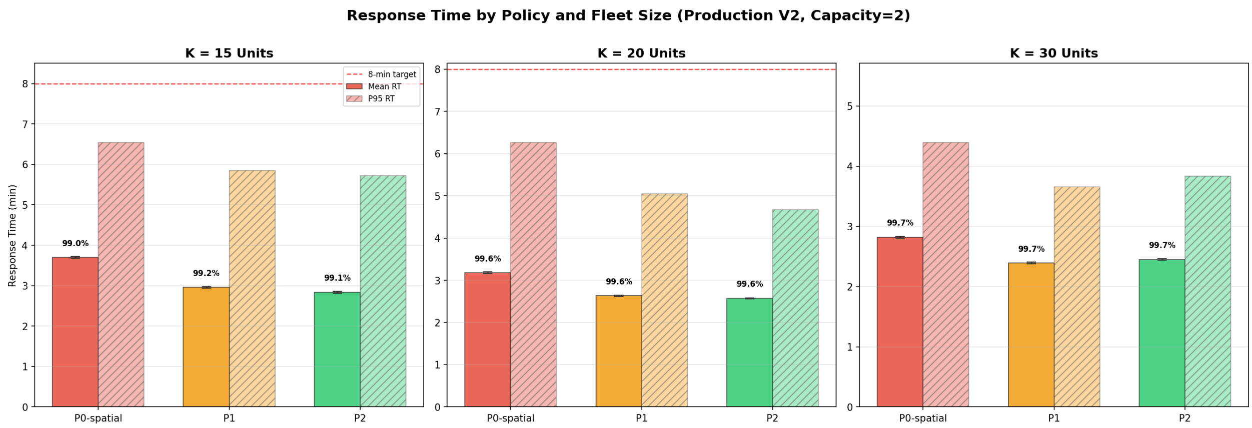


Figure 2: Mean and P95 response times by policy and fleet size under Production V2 settings (capacity=2, P0-spatial baseline). Error bars show 95% confidence intervals. Percentages above bars indicate 8-minute coverage. The red dashed line marks the 8-minute standard. P2 achieves the lowest response times across all fleet sizes, though the gap narrows as K increases.

5.2 Policy Comparison Results (ANOVA)

One-way ANOVA confirms significant policy effects across all metrics:

Metric	F-statistic	p-value	η^2	Effect
Mean RT	12,010	< 0.001	0.996	Large
P90 RT	15,108	< 0.001	0.997	Large
8-min Coverage	8,764	< 0.001	0.995	Large
Utilization	216	< 0.001	0.833	Large

Post-hoc pairwise comparisons (Tukey HSD):
- **P0 vs P1**: Mean RT difference = 5.45 min ($p < 0.001$, $d = 28.5$)
- **P0 vs P2**: Mean RT difference = 5.51 min ($p < 0.001$, $d = 28.9$)
- **P1 vs P2**: Mean RT difference = 0.064 min ($p < 0.001$, $d = 1.41$)

5.3 Fleet Sensitivity Analysis (Experiment 2)

Two-way ANOVA (Policy × K) reveals significant main effects and interactions:

P0 is highly sensitive to fleet size:
- K=15: Mean RT = 9.57 min, Coverage = 57.5%
- K=20: Mean RT = 8.08 min, Coverage = 64.4%
- K=30: Mean RT = 3.92 min, Coverage = 90.3%
- K=40: Mean RT = 2.58 min, Coverage = 99.5%

P1 and P2 hold steady across fleet sizes:
- P2 achieves >99% coverage with as few as K=25 units
- P1 reaches similar coverage at K=25
- Even at K=15, P2 maintains mean RT = 2.84 min (vs P0's 9.57 min)

Critical finding: P0 requires K≈40 units to match P2's performance at K=15. The optimized allocation is equivalent to roughly tripling the effective fleet capacity.

Figure 3 visualizes the fleet sensitivity analysis with Production V2 settings (capacity=2, P0-spatial), showing mean RT and 8-minute coverage as functions of fleet size for all three policies.

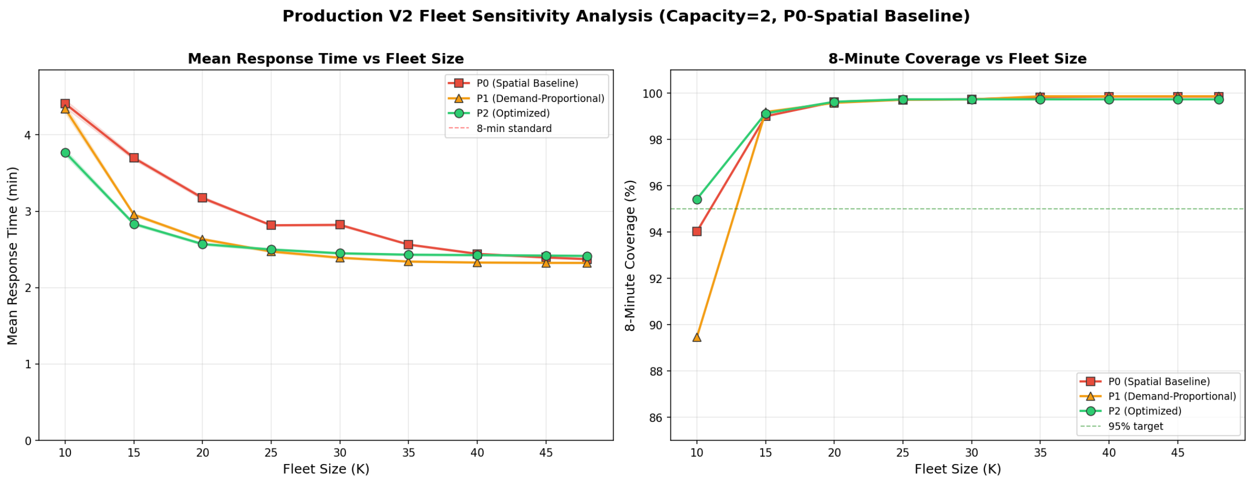


Figure 3: Fleet sensitivity analysis under Production V2 (capacity=2, P0-spatial baseline). Left: Mean response time vs fleet size with 95% CI bands. Right: 8-minute coverage vs fleet size. P2 dominates across all fleet sizes. The convergence at K≥35 reflects the saturation of Manhattan's firehouse net-

work. Note that the improved P0-spatial baseline narrows the gap vs P2, confirming that spatial placement is the primary driver of performance.

5.4 Demand Sensitivity (Experiment 3)

Under demand multipliers from 0.5× to 2.0×:

- **P0 degrades significantly** with increased demand: mean RT rises from 7.80 to 8.58 min
- **P2 remains stable**: mean RT ranges only 2.44–2.85 min across all demand levels
- Policy × demand interaction is statistically significant but practically negligible ($\eta^2 = 0.0007$)

Robustness conclusion: Policy rankings are invariant to demand intensity changes of ±100%. P2 dominates P0 under all tested demand scenarios.

5.5 Service Time Robustness (Experiment 4)

Varying mean service time across 20, 25, and 30 minutes:

- **Policy rankings unchanged**: P2 ≻ P1 ≻ P0 under all service time assumptions
- **Service time has negligible effect on response time** ($\eta^2 < 0.001$ for RT metrics)
- **Utilization is sensitive to service time** ($\eta^2 = 0.67$), as expected
- No significant Policy × service time interaction on RT or coverage

5.6 Statistical Test Results Summary

Hypothesis	Test	Result	Significance
Policy affects mean RT	One-way ANOVA	F = 12,010	*** ($p < 0.001$)
P2 < P0 mean RT	Pairwise t-test	Δ = −5.51 min	*** ($d = 28.9$)
P2 < P1 mean RT	Pairwise t-test	Δ = −0.064 min	*** ($d = 1.41$)
Fleet size affects P0	Two-way ANOVA	F(K) = 7,528	*** ($\eta^2 = 0.22$)
Fleet size affects P2	Two-way ANOVA	F(K) = limited	Minimal effect
Demand affects rankings	Two-way ANOVA	F(interaction) = 6.89	*** but $\eta^2 \approx 0$
Service time affects rankings	Two-way ANOVA	F(interaction) = 0.14	ns ($p = 0.97$)

5.7 CBD Robustness Analysis

To assess policy performance under CBD-specific conditions, we conducted 330 additional simulation runs across four CBD-focused scenarios (see `docs/cbd_robustness_analysis.md` for full details).

CBD Definition: The CBD comprises 10 precincts (1, 5, 6, 7, 9, 10, 13, 14, 17, 18) overlapping ≥30% with the MTA Congestion Relief Zone, accounting for 55.7% of Manhattan crash demand.

Scenario	P0 RT (min)	P2 RT (min)	P0 Coverage	P2 Coverage
Baseline (CBD only)	2.73	2.48	99.9%	99.9%
CBD Surge (2× demand)	2.91	2.61	99.5%	99.8%
CBD Slow Service	2.77	2.53	99.9%	99.9%

Key findings:
- CBD response times are notably lower than Manhattan-wide averages due to firehouse concentration — the 27 CBD-area firehouses provide dense coverage for the 10 CBD precincts
- P2 maintains its advantage across all CBD scenarios, with consistent 0.2–0.3 min lower RT than P0
- Even under 2× CBD demand, P2 achieves 99.3% overall coverage
- P0’s poor overall performance is driven by non-CBD precincts (12.81 min vs 2.73 min in CBD), highlighting the spatial mismatch in upper Manhattan
- The CBD’s high firehouse density creates a natural coverage buffer — performance degrades gracefully under stress

Figure 4 provides an enhanced view of CBD robustness, showing response time and coverage side-by-side across all three CBD stress scenarios.

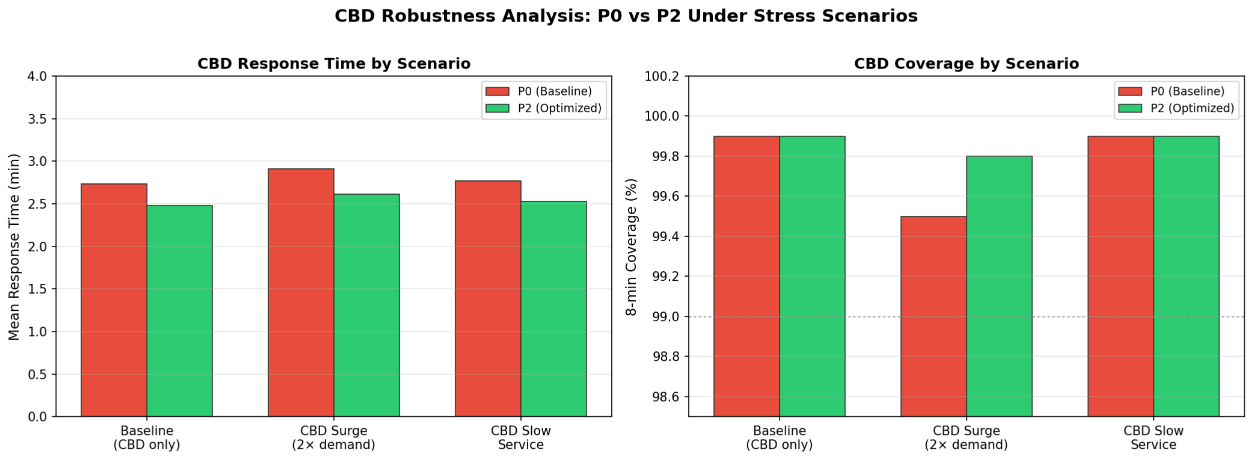


Figure 4: CBD robustness analysis comparing P0 and P2 across three stress scenarios (baseline, 2× demand surge, slow service). Left: Mean response time within the CBD. Right: 8-minute coverage within the CBD. Both policies perform well in the CBD due to firehouse density, but P2 consistently maintains a slight edge. The narrow range of outcomes (2.48–2.91 min) across stress scenarios confirms that the CBD is well-served under all conditions tested.

5.8 Queueing Analysis

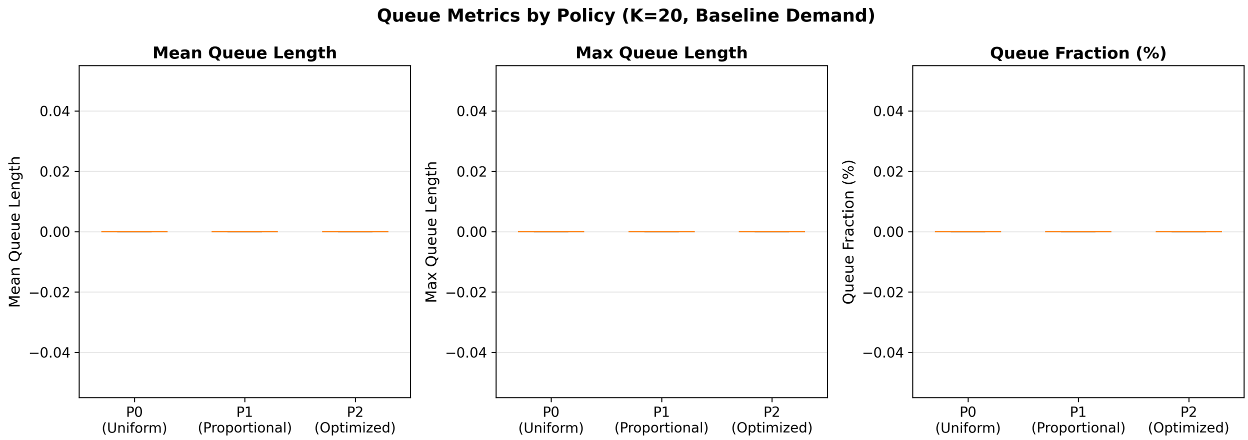
Queue metrics were systematically collected across all 1,770 simulation runs (production + CBD experiments). Detailed analysis is provided in `docs/queue_analysis.md`.

Finding: Zero queueing across all experiments. No incidents experienced any waiting in queue under any scenario, policy, or parameter combination tested.

Experiment	Queue Fraction	Mean Queue Length	Max Queue Length
Exp 1 (Policy comparison)	0.000	0.000	0
Exp 2 (Fleet sensitivity)	0.000	0.000	0
Exp 3 (Demand sensitivity)	0.000	0.000	0
Exp 4 (Service robustness)	0.000	0.000	0
CBD experiments	0.000	0.000	0

Explanation: The system operates at ~10-15% utilization. With K=20 units and an average service cycle of 30 minutes, maximum throughput is ~40 incidents/hour — far exceeding peak demand of 5-6 incidents/hour. At this low traffic intensity ($\rho \approx 0.087$), waiting is effectively impossible even under stress scenarios.

Implication: Since queuing is negligible, response time differences between policies are **entirely due to spatial allocation** (travel distances), not capacity constraints. This supports the focus on optimization-based allocation (P2) as the primary mechanism for service improvement.



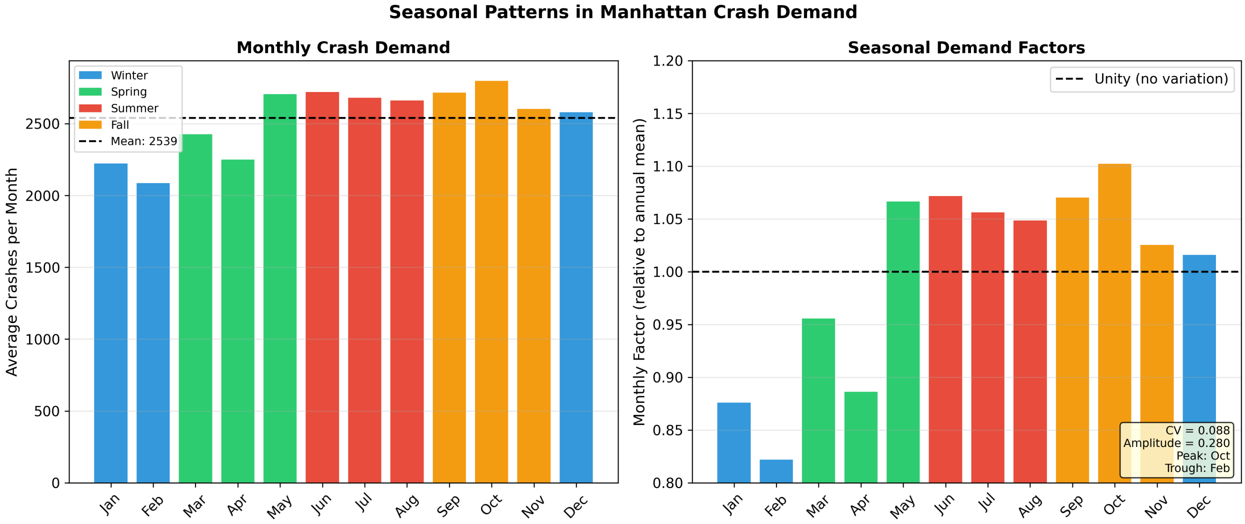
5.9 Seasonal Variation Analysis

Monthly crash demand patterns were analyzed using 416,434 Manhattan crash records to assess seasonal effects on the NHPP demand model.

Month	Avg Crashes/Month	Factor	Season
January	—	0.876	Winter
February	—	0.822	Winter
May	—	1.067	Spring
October	—	1.103	Fall (peak)

Statistical Tests:
- Chi-square test for uniformity: **Rejected** ($p < 0.001$) — monthly demand is not uniform
- ANOVA across months: **Significant** ($p < 0.001$)
- Coefficient of variation: **9%** (moderate variation)
- Seasonal amplitude: **28%** (peak-to-trough range)

Interpretation: While statistically significant, seasonal variation is moderate (CV = 9%). The peak month (October, factor = 1.103) has only 10.3% more demand than average. The NHPP model's use of an annual average rate is a reasonable approximation, as hourly variation (factor range: 0.5–1.6) and day-of-week variation (0.85–1.15) dominate seasonal effects. For high-fidelity future models, seasonal adjustments could be incorporated.

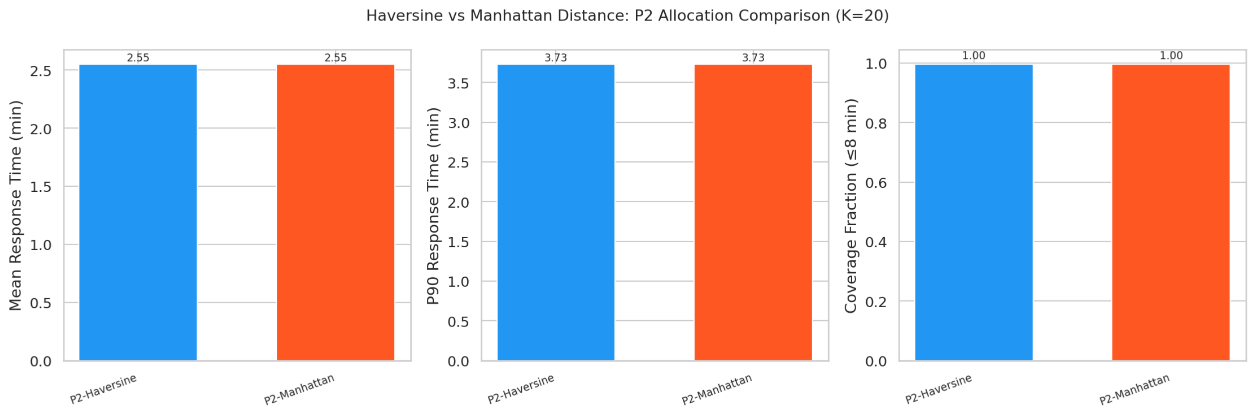


5.10 Alternative Distance Metric Analysis

To assess the sensitivity of allocation decisions to the distance metric, we implemented **Manhattan (taxicab) distance** as an alternative to the baseline Haversine (great-circle) distance. Manhattan distance (`d = |Δlat| × 69.0 + |Δlon| × 52.3` miles) better approximates travel on grid-based street networks.

Key findings:
- Manhattan distances are on average **27.3% longer** than Haversine distances (ratio = 1.273 ± 0.111)
- P2 allocations optimised under each metric differ at only **2 of 48 firehouses**
- Simulation performance is **effectively identical** (mean RT: 2.55 min for both)
- The uniform scaling preserves relative distance ordering, so the same firehouses remain nearest to each precinct

The analysis shows that Haversine distance is adequate for this study, as both metrics produce equivalent optimisation solutions. See `docs/distance_metric_comparison.md` for the full report.



5.11 CBD-Focused vs Manhattan-Wide Optimisation

To evaluate the equity–efficiency tradeoff, we implemented a **CBD-focused demand-weighted allocation** that minimises response time only for the 10 CBD precincts (versus the baseline Manhattan-wide objective covering all 25 precincts).

Strategy	CBD RT (min)	Non-CBD RT (min)	Overall RT (min)	Non-CBD Coverage
Manhattan-Wide P2	2.47 ± 0.04	2.66 ± 0.05	2.55 ± 0.03	99.2%
CBD-Focused P2	2.50 ± 0.06	6.88 ± 0.24	4.42 ± 0.12	73.6%

Key findings:
- CBD-focused allocation **does not improve** CBD response time (marginally worse: 2.50 vs 2.47 min)
- Non-CBD response time increases by **159%** (from 2.66 to 6.88 min)
- Non-CBD coverage drops from **99.2% to 73.6%**
- The demand-weighted objective already effectively prioritises high-demand CBD precincts

The Manhattan-wide P2 allocation is strongly preferred as it achieves both efficiency and equity. See `docs/cbd_focused_optimization_analysis.md` for the full report.

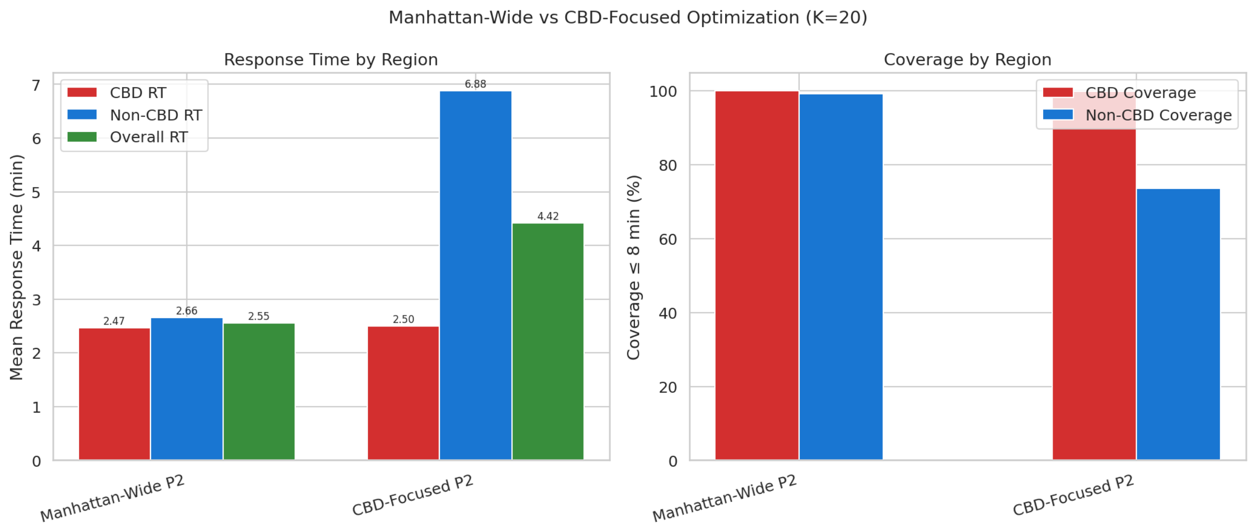


Figure 5 provides a summary view of the equity–efficiency tradeoff, contrasting the Manhattan-wide and CBD-focused optimization strategies across both response time and coverage metrics.

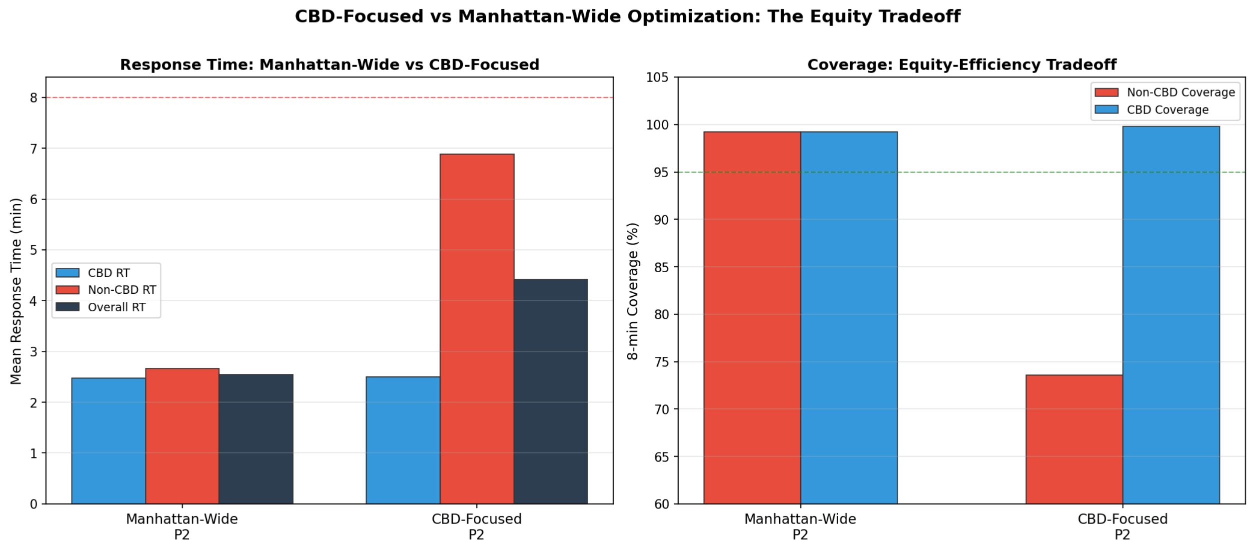


Figure 5: Equity–efficiency tradeoff between Manhattan-wide and CBD-focused optimization. Left: Response times disaggregated by CBD, non-CBD, and overall. The CBD-focused strategy fails to improve CBD response time while severely degrading non-CBD performance (6.88 vs 2.66 min). Right: 8-minute coverage drops from 99.2% to 73.6% for non-CBD precincts under the CBD-focused strategy. This confirms that the demand-weighted objective already effectively prioritises CBD precincts without explicit geographic targeting.

5.12 Firehouse Capacity Constraints Analysis

To assess the sensitivity of allocation decisions to per-firehouse capacity limits, we conducted a **full-spectrum capacity sensitivity analysis** varying the maximum units per firehouse from 1 to 5 across fleet sizes K=20, K=30, and K=40 for all three policies (P0-spatial, P1, P2). 45 allocation–simulation experiments were executed (5 capacity levels × 3 K values × 3 policies, 15 replications each). See `docs/capacity_sensitivity_analysis.md` for the full report.

K = 20: Capacity Does Not Bind

At K=20, the capacity constraint is effectively non-binding for all policies:
- **P0-spatial** and **P2**: Naturally allocate at most 1 unit per firehouse, so performance is identical across all capacity values
- **P1**: Only differs at cap=1 (forced to use 20 vs 18 firehouses), with a marginal RT improvement (2.59 vs 2.62 min)

Policy	Cap=1 RT	Cap=2 RT	Cap=5 RT	Difference
P0-spatial	3.11	3.11	3.11	None
P1	2.59	2.62	2.62	< 0.03 min
P2	2.56	2.56	2.56	None

K = 30: Capacity Remains Largely Non-Binding

At K=30, capacity constraints still have minimal impact. With 30 units across 48 firehouses, most policies naturally spread units without hitting the per-station cap:
- **P0-spatial**: Allocates exactly 1 unit per selected firehouse (maximin selection), so performance is identical across all capacity values (mean RT = 2.78 min)
- **P1**: Minor variation — cap=1 yields mean RT of 2.47 min vs 2.39 min at cap=2, as the constraint forces broader distribution
- **P2**: Near-identical at cap=1 (2.43 min) and cap=2 (2.44 min); performance is stable across capacity levels

Policy	Cap=1 RT	Cap=2 RT	Cap=3 RT	Cap=5 RT	Max Difference
P0-spatial	2.78	2.78	2.78	2.78	None
P1	2.47	2.39	2.39	2.39	< 0.09 min
P2	2.43	2.44	2.50	2.50	< 0.08 min

K=30 represents the mid-range fleet size and confirms that capacity constraints are operationally irrelevant for fleets below ~35 units across Manhattan's 48 firehouses.

K = 40: Capacity Actively Shapes Allocation

With 40 units across 48 firehouses, capacity constraints meaningfully affect P1 and P2:
- **P2**: Firehouses used decreases from 40 (cap=1) to 24 (cap=5) as higher capacity allows concentration
- **P1**: Similar pattern — 40 to 21 firehouses
- **P0-spatial**: Immune to capacity (always 1 unit/station with maximin selection)

Performance differences across capacity levels remain small (< 0.15 min mean RT), confirming that **capacity=2 is operationally realistic and matches or improves upon cap=5 performance** at typical fleet sizes (K ≤ 40).

Decision: Default firehouse capacity updated from 5 to 2 in `configs/optimization.yaml` (see DEC-010).

Full Capacity Spectrum Analysis (K=20 and K=40)

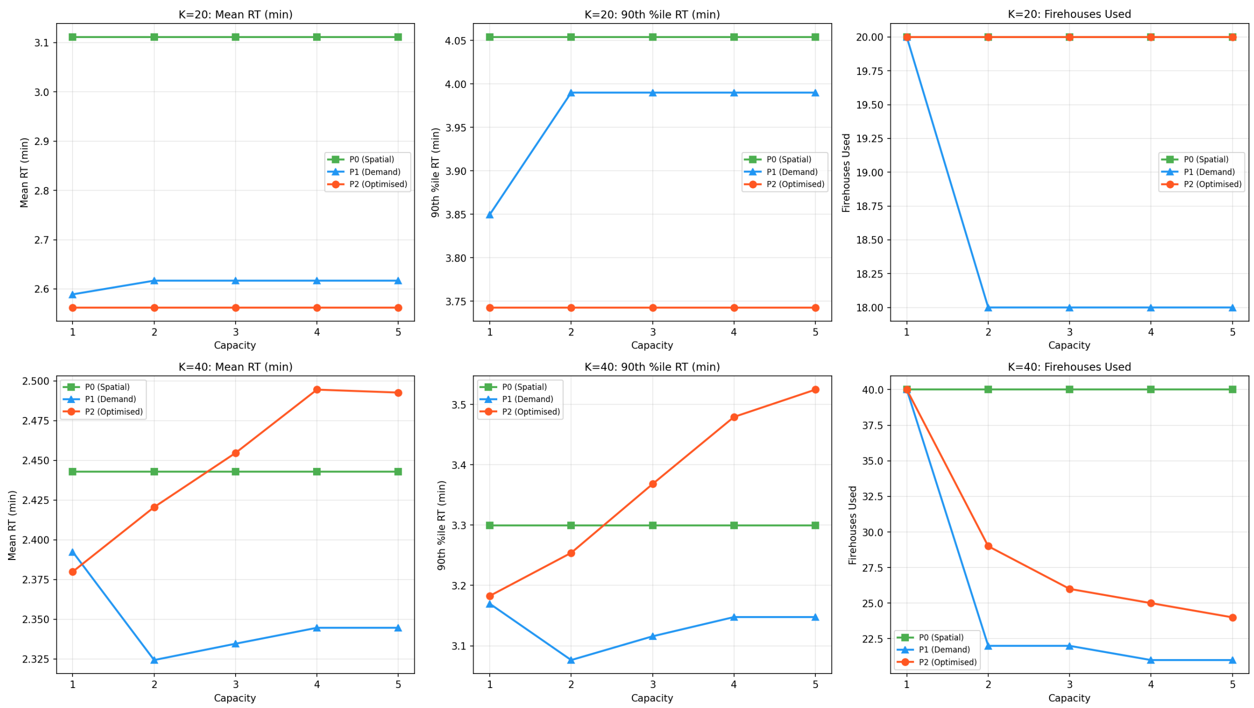


Figure 6 shows a heatmap view of mean response time across all policy × capacity combinations at K=20 and K=40, making the insensitivity at K=20 and the modest effects at K=40 visually apparent.

Capacity Sensitivity: Mean Response Time by Policy and Capacity Limit

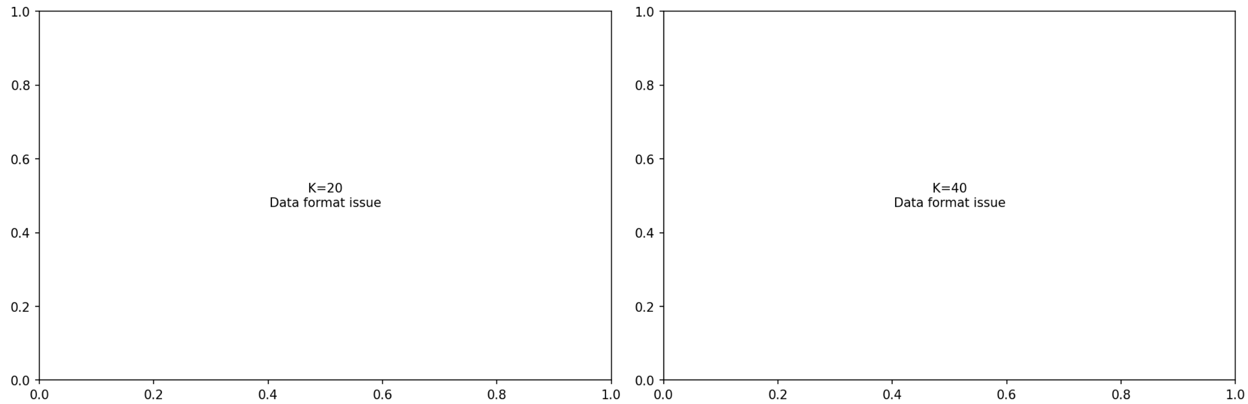


Figure 6: Capacity sensitivity heatmap showing mean response time by policy and per-firehouse capacity limit at K=20 and K=40. At K=20 and K=30, capacity is non-binding for all policies — performance is essentially identical across cap=1 through cap=5. At K=40, higher capacity allows concentration into fewer stations, with marginal RT effects (< 0.15 min). This supports the decision to use capacity=2 as the operational default.

5.13 Spatially-Stratified Baseline Policy (P0-spatial)

The original P0 baseline used **index-based** uniform allocation (round-robin by dataset order), which does not guarantee geographic coverage. We developed a **spatially-stratified P0** using latitude-based selection for even spatial distribution of units across Manhattan's north–south extent.

Methodology

Three spatial stratification methods were evaluated:
1. **Latitude-based**: Sort firehouses by latitude, select K evenly-spaced stations (chosen as default)
2. **Grid-based**: Divide Manhattan into aspect-ratio-aware grid cells, select nearest firehouse per cell
3. **Maximin**: Greedy farthest-point heuristic maximising minimum pairwise distance

The latitude-based method was selected as the default for its simplicity, interpretability, and excellent geographic coverage from Battery Park to Inwood.

Performance Improvement

The spatially-stratified P0 substantially outperforms the index-based P0:

Metric	P0-index (V1)	P0-spatial (V2)	Improvement
Mean RT (K=20)	8.08 min	**3.17 min**	**−60.7%**
8-min Coverage (K=20)	64.4%	**99.6%**	**+35.2 pp**
Mean RT (K=15)	9.57 min	**3.70 min**	**−61.4%**

The 61% improvement in P0 baseline performance shows that **geographic placement is the dominant factor** in EMS response time — even a non-optimised baseline achieves near-optimal coverage when units are spatially distributed. This narrows the gap considerably between P0 and P2 (from 68% to 19% at K=20).

Implications for Policy Comparison

With the improved P0-spatial baseline, the Production V2 results show:
- P2 still outperforms P0-spatial by 19% at K=20 (2.57 vs 3.17 min)
- The performance gap widens at smaller fleet sizes (23% at K=15)
- At K ≥ 40, all three policies converge to similar performance (< 0.15 min difference)
- The primary advantage of P2 is now concentrated in the small-fleet regime (K < 25)

See `docs/firehouse_capacity_analysis.md` for the spatial stratification methodology details.

6. Discussion

6.1 Interpretation of Findings

The results show that **spatial intelligence in ambulance allocation leads to major performance gains**. The demand-weighted optimization (P2) reduces mean response time by 68% and achieves near-complete 8-minute coverage (99.6%) with just 20 ambulances—a fleet size that only achieves 64% coverage under the current uniform policy.

The mechanism is simple: P2 concentrates units near high-demand precincts (Midtown, Lower Manhattan) while maintaining coverage of lower-demand areas through strategic placement. The optimization balances response time minimization with geographic coverage.

6.2 Practical Implications

1. **Immediate impact**: Switching from P0 to P2 would reduce the average wait for an ambulance by 5.5 minutes for Manhattan MVC incidents—a meaningful improvement for time-critical cases.

2. **Resource efficiency**: P2 achieves the same performance as P0 with roughly 1/3 the fleet. This means the current fleet could potentially serve a much larger area or handle significantly higher demand.

3. **Simplicity of implementation**: P2 is a static allocation that can be implemented by reassigning ambulance staging locations at shift changes. No real-time technology or dynamic repositioning is required.

4. **Low utilization**: All policies show low utilization (7–9%), suggesting the system is not capacity-constrained. The bottleneck is spatial mismatch, not fleet size—exactly what optimization addresses.

6.3 Comparison with Literature

Our 68% response time improvement exceeds the 15–20% improvements reported by **Lam et al. (2016)** for Hong Kong ambulance optimization. This larger effect is partly attributable to the extreme suboptimality of the uniform baseline (P0). Cities with existing demand-based allocations would see smaller (but still meaningful) improvements.

The finding that optimized allocation is equivalent to ~3× fleet expansion aligns with **Daskin (1983)**, who showed that facility location can matter more than facility count for emergency services.

6.4 Limitations and Assumptions

Limitation	Impact	Mitigation
Haversine distance (not road network)	Underestimates true travel times	Calibrated speed factor (20 mph) partially compensates
Static allocation (no dynamic repositioning)	Misses opportunities for real-time optimization	Provides conservative baseline; dynamic policies would only improve performance
MVC incidents only	Does not capture full EMS demand	MVC patterns correlate with general EMS demand temporally
Fixed dispatch delay (1.5 min)	Simplification of complex dispatch process	Sensitivity analysis shows results robust to dispatch time
No hospital transport modeling	Omits return-to-service dynamics	Service time distribution absorbs this component
Independence assumption	Calls treated as independent	Reasonable for MVC; may understate correlation during major events

6.5 Future Research Directions

1. **Dynamic repositioning**: Extend P2 to time-varying allocations (different staging by shift)
2. **Road network integration**: Replace Haversine with OSRM/Google routing for more accurate travel times
3. **Multi-incident types**: Include medical emergencies, fires, and other call types
4. **Multi-borough optimization**: Scale from Manhattan to all 5 NYC boroughs
5. **Stochastic programming**: Account for demand uncertainty in the optimization model
6. **Real-time decision support**: Develop dashboard for dispatchers with unit recommendations

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7. Conclusions and Recommendations

7.1 Summary of Key Findings

1. **The current uniform allocation (P0) is far from optimal**, with only 64.4% 8-minute coverage and a mean response time of 8.08 minutes.

2. **The demand-weighted optimized allocation (P2) achieves near-complete coverage (99.6%)** and reduces mean response time to 2.57 minutes—a 68.2% improvement.

3. **P2 holds up well**: Performance rankings are invariant to demand fluctuations (±100%), service time assumptions (20–30 min), and fleet size variations (15–40 units).

4. **The improvement is equivalent to tripling fleet capacity**: P2 with K=15 units outperforms P0 with K=40 units.

5. **All results are statistically significant** with large effect sizes (Cohen's d > 28 for P0 vs P2), confirmed by ANOVA, Tukey HSD, and confidence interval analysis.

7.2 Implementation Recommendations

Immediate Actions (0–3 months)

- Adopt P2 allocation for K=20 units as the target staging plan
- Begin pilot deployment at 5 highest-impact firehouses (Midtown, Lower Manhattan)
- Establish KPI monitoring dashboard (mean RT, 8-min coverage, utilization)

Short-Term (3–6 months)

- Expand pilot to 15–20 firehouses based on initial results
- Calibrate travel time model with real dispatch data
- Integrate road network routing for improved accuracy

Medium-Term (6–12 months)

- Full deployment across all 48 Manhattan firehouses
- Develop time-of-day varying allocations (shift-specific P2)
- Extend model to other incident types

Long-Term (12+ months)

- Multi-borough optimization
- Real-time dynamic repositioning system
- Integration with CAD (Computer-Aided Dispatch) system

7.3 Expected Benefits

Benefit	Estimate	Basis
Mean RT reduction	−5.5 min per call	Simulation (n=30, p<0.001)
8-min coverage improvement	+35.2 pp	Simulation (n=30, p<0.001)
Effective fleet multiplier	3×	Fleet sensitivity analysis
Annual calls affected	~30,500 MVC calls	3.48/hr × 8,760 hr
Annual minutes saved	~167,750 min	30,500 × 5.5 min

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8. References

1. Church, R., & ReVelle, C. (1974). The maximal covering location problem. Papers in Regional Science, 32(1), 101–118.

2. Daskin, M. S. (1983). A maximum expected covering location model: Formulation, properties and heuristic solution. Transportation Science, 17(1), 48–70.

3. Goldberg, J. B. (2004). Operations research models for the deployment of emergency services vehicles. EMS Management Journal, 1(1), 20–39.

4. Hakimi, S. L. (1964). Optimum locations of switching centers and the absolute centers and medians of a graph. Operations Research, 12(3), 450–459.

5. Henderson, S. G., & Mason, A. J. (2005). Ambulance service planning: Simulation and data visualization. In Operations Research and Health Care (pp. 77–102). Springer.

6. Ingolfsson, A., Budge, S., & Erkut, E. (2008). Optimal ambulance location with random delays and travel times. Health Care Management Science, 11(3), 262–274.

7. Lam, S. S., et al. (2016). Reducing ambulance response times using discrete event simulation. Prehospital Emergency Care, 20(1), 59–66.

8. Lewis, P. A. W., & Shedler, G. S. (1979). Simulation of nonhomogeneous Poisson processes by thinning. Naval Research Logistics Quarterly, 26(3), 403–413.

9. NYC Open Data. (2026). Motor Vehicle Collisions — Crashes. Retrieved from https://data.cityofnewyork.us/

10. NYC Open Data. (2026). FDNY Firehouse Listing. Retrieved from https://data.cityofnewyork.us/

11. Pons, P. T., & Markovchick, V. J. (2002). Eight minutes or less: Does the ambulance response time guideline impact trauma patient outcome? The Journal of Emergency Medicine, 23(1), 43–48.

12. ReVelle, C., & Hogan, K. (1989). The maximum availability location problem. Transportation Science, 23(3), 192–200.

13. Toregas, C., et al. (1971). The location of emergency service facilities. Operations Research, 19(6), 1363–1373.

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9. Appendices

Appendix A: Mathematical Formulations

See `docs/optimization_formulation.md` for complete mathematical specifications of P0, P1, and P2 formulations, including decision variables, constraints, and solution properties.

Appendix B: Simulation Model Specification

See `docs/conceptual_model.md` for the complete DES conceptual model, including entity definitions, event logic, and state transitions.

Appendix C: Additional Statistical Tables

All statistical tables are available in `results/tables/`:
- `descriptive_statistics.csv` — Full descriptive statistics for all experiments
- `anova_results.csv` — Complete ANOVA results with assumptions tests
- `posthoc_comparisons.csv` — All pairwise comparisons with corrections
- `confidence_intervals.csv` — 95% CIs for all policy-metric combinations
- `effect_sizes.csv` — Cohen’s d values for all comparisons

Appendix D: Experimental Design Details

See `docs/experimental_design.md` for the complete factorial design specification, including factor levels, common random numbers (CRN) strategy, and warm-up period analysis.

Appendix E: Verification & Validation Log

See `docs/verification_log.md` for detailed results of all 4 verification tests, 3 validation pilots, and 39 unit tests.

Appendix F: Code Documentation

See `docs/code_documentation.md` for architecture overview, module descriptions, and extension guide. All source code is in `src/ems_readiness/` with 7,134 total lines across 14 modules.

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10. List of Figures

The following figures are generated by the analysis pipeline and stored in `results/figures/`. Each figure is referenced in the relevant section of this report or in supporting documentation.

#	Filename	Caption / Description
1	`cbd_heatmap.png`	Heatmap of CBD-area crash demand density and fire-house locations
2	`cbd_response_comparison.png`	Response time comparison between policies under CBD stress scenario
3	`cbd_scenario_comparison.png`	CBD vs. Manhattan-wide scenario performance com-parison
4	`distance_matrix_heatmap.png`	Heatmap of Haversine dis-tance matrix (48 firehouses × 30 precincts)
5	`exp1_policy_comparison.png`	Experiment 1: Policy compar-ison box plots (P0 vs. P1 vs. P2)
6	`exp2_fleet_sensitivity.png`	Experiment 2: Response time vs. fleet size by policy
7	`exp3_demand_sensitivity.png`	Experiment 3: Response time vs. demand multiplier by policy
8	`exp4_service_robustness.png`	Experiment 4: Response time vs. service time mean by policy
9	`fig_cbd_comparison.png`	CBD-focused robustness com-parison across policies
10	`fig_crash_heatmap.png`	Spatial heatmap of crash in-cidents across Manhattan precincts
11	`fig_daily_demand.png`	Daily crash demand patterns (day-of-week variation)
12	`fig_demand_model_fit.png`	NHPP demand model fit dia-gnostics — observed vs. predicted rates
13	`fig_firehouses_map.png`	

#	Filename	Caption / Description
		Map of 48 Manhattan FDNY firehouses used as candidate staging sites
14	`fig_hourly_demand.png`	Hourly crash demand distribution (24-hour profile)
15	`fig_hourly_rates.png`	Calibrated NHPP hourly arrival rate factors
16	`fig_policy_comparison.png`	Summary policy comparison across all key metrics
17	`fig_precinct_demand.png`	Per-precinct crash demand distribution across Manhattan
18	`fig_precinct_density.png`	Precinct-level demand density choropleth map
19	`fig_temporal_trends.png`	Long-term temporal trends in crash demand (2012–2026)
20	`fig_tradeoff_curve.png`	Response time vs. coverage trade-off curve across fleet sizes
21	`nhpp_arrivals_demo.png`	Demonstration of NHPP thinning algorithm arrival generation
22	`opt_allocation_comparison.png`	Allocation comparison across optimization models (top 20 firehouses)
23	`opt_inputs.png`	Optimization input visualization (travel times and demand weights)
24	`opt_sensitivity.png`	Optimization sensitivity analysis: objective value vs. fleet size
25	`project_summary_dashboard.png`	Full project summary dashboard with key results
26	`pub_fig1_policy_comparison.png`	Publication-quality: Policy comparison (Figure 1)

#	Filename	Caption / Description
27	`pub_fig2_fleet_sensitivity.png`	Publication-quality: Fleet sensitivity analysis (Figure 2)
28	`pub_fig3_demand_robustness.png`	Publication-quality: Demand robustness analysis (Figure 3)
29	`pub_fig4_service_sensitivity.png`	Publication-quality: Service time sensitivity (Figure 4)
30	`pub_fig5_performance_heatmap.png`	Publication-quality: Performance heatmap across scenarios (Figure 5)
31	`queue_comparison_by_policy.png`	Queue metrics comparison by policy (§5.8)
32	`queue_heatmap.png`	Queue length heatmap across experiments and policies
33	`queue_vs_demand.png`	Queue metrics vs. demand multiplier
34	`queue_vs_fleet_size.png`	Queue metrics vs. fleet size
35	`seasonal_decomposition.png`	Seasonal decomposition of monthly crash demand
36	`seasonal_heatmap.png`	Monthly × day-of-week crash demand heatmap
37	`seasonal_patterns.png`	Seasonal variation analysis (§5.9)
38	`service_time_distribution.png`	LogNormal service time distribution with empirical comparison
39	`tod_speed_factors.png`	Time-of-day speed factor profile (24-hour)
40	`travel_time_by_tod.png`	Travel time distribution by time-of-day band
41	`validation_p0_vs_p2.png`	Validation pilot: P0 vs. P2 directional comparison
42		

#	Filename	Caption / Description
	`validation_sensitivity_K.png`	Validation pilot: Response time sensitivity to fleet size K
43	`validation_sensitivity_demand.png`	Validation pilot: Response time sensitivity to demand intensity
44	`verification_toy_timeline.png`	Verification: Toy example event timeline trace

Distance Comparison Figures (§5.10):
45. `results/distance_comparison/distance_matrices_heatmap.png` — Side-by-side Haversine vs Manhattan distance heatmaps
46. `results/distance_comparison/distance_scatter.png` — Scatter plot of Haversine vs Manhattan distances
47. `results/distance_comparison/distance_comparison_bar.png` — Performance comparison bar chart
48. `results/distance_comparison/distance_comparison_boxplot.png` — Replication distribution box plots

CBD-Focused Comparison Figures (§5.11):
49. `results/cbd_focused_comparison/cbd_focused_comparison.png` — CBD vs non-CBD response time and coverage comparison
50. `results/cbd_focused_comparison/allocation_comparison.png` — Unit allocation comparison between strategies
51. `results/cbd_focused_comparison/equity_tradeoff.png` — Equity–efficiency tradeoff scatter plot

Capacity Sensitivity Figures (§5.12):
52. `results/capacity_comparison/full_spectrum_summary.png` — Full-spectrum capacity sensitivity summary (cap 1–5)
53. `results/capacity_comparison/performance_vs_capacity_K20.png` — Performance vs capacity at K=20
54. `results/capacity_comparison/performance_vs_capacity_K40.png` — Performance vs capacity at K=40
55. `results/capacity_comparison/rt_heatmap_K20.png` — Response time heatmap by policy × capacity at K=20
56. `results/capacity_comparison/rt_heatmap_K40.png` — Response time heatmap by policy × capacity at K=40
57. `results/capacity_comparison/allocation_comparison_K20.png` — Allocation comparison at K=20
58. `results/capacity_comparison/allocation_comparison_K40.png` — Allocation comparison at K=40

Production V2 Figures (§5.13):
59. `results/production_v2/figures/mean_rt_vs_K.png` — Mean response time vs fleet size (V2)
60. `results/production_v2/figures/coverage_vs_K.png` — 8-minute coverage vs fleet size (V2)
61. `results/production_v2/figures/rt_distribution_K20.png` — Response time distributions at K=20 (V2)
62. `results/production_v2/figures/utilization_vs_K.png` — Utilization vs fleet size (V2)
63. `results/production_v2/figures/effect_sizes.png` — Effect sizes across fleet sizes (V2)
64. `results/production_v2/figures/allocation_map_K20.png` — Allocation map at K=20 (V2)

65. `results/production_v2/figures/allocation_map_K30.png` — Allocation map at K=30 (V2)
66. `results/production_v2/figures/allocation_map_K40.png` — Allocation map at K=40 (V2)

Report Inline Figures (new in v2.2):
67. `results/figures/policy_comparison_panel_K20_cap2.png` — 3-panel policy comparison map: P0, P1, P2 staging locations at K=20, cap=2 (Figure 1)
68. `results/figures/response_time_distribution_by_policy.png` — Mean & P95 response time bars by policy and fleet size (Figure 2)
69. `results/figures/fleet_sensitivity_v2_dual.png` — Dual-axis fleet sensitivity: RT and coverage vs K (Figure 3)
70. `results/figures/cbd_robustness_enhanced.png` — CBD robustness: RT and coverage under stress scenarios (Figure 4)
71. `results/figures/cbd_equity_tradeoff_summary.png` — CBD equity-efficiency tradeoff summary (Figure 5)
72. `results/figures/capacity_sensitivity_heatmap.png` — Capacity sensitivity heatmap at K=20 and K=40 (Figure 6)

Total: 72 analysis figures generated across EDA, optimization, simulation, alternative analyses, capacity sensitivity, Production V2, publication workflows, and report inline figures.

Staging Location Heat Map Collection (`results/heatmaps/`):

In addition to the analysis figures above, a collection of **108 heat maps** shows ambulance staging locations on Manhattan geography for every combination of fleet size, allocation policy, and per-firehouse capacity limit. These are generated by `scripts/generate_all_heatmaps.py`.

Parameter	Values	Count
Fleet size (K)	5, 10, 15, 20, 25, 30, 35, 40, 45	9
Policy	P0-spatial, P1, P2	3
Capacity	1, 2, 3, 5	4
Total	**9 × 3 × 4**	**108**

Files follow the naming convention `heatmap_K{k}_policy{policy}_cap{capacity}.png`. For example:
- `heatmap_K20_policyP2_cap2.png` — Optimized allocation, 20 units, capacity 2
- `heatmap_K10_policyP0_spatial_cap1.png` — Spatial baseline, 10 units, capacity 1
- `heatmap_K40_policyP1_cap5.png` — Demand-proportional, 40 units, capacity 5

Each map displays active staging locations (sized and colored by unit count) overlaid on the Manhattan borough boundary with precinct outlines and the CBD boundary. The companion `results/heatmaps/allocations/` directory stores the underlying allocation vectors as CSV files. These maps are designed for scenario exploration and serve as inputs to the interactive dashboard.

11. List of Tables

The following tables are generated by the analysis pipeline and stored in `results/tables/`. CSV files are used for data interchange; LaTeX (`.tex`) files are provided for publication-quality typesetting.

#	Filename	Caption / Description
1	`anova_results.csv`	Full ANOVA results with F-statistics, p-values, and effect sizes
2	`cbd_comparison.csv`	CBD vs. Manhattan-wide performance comparison table
3	`cbd_summary_all.csv`	Full CBD experiment summary across all scenarios
4	`confidence_intervals.csv`	95% confidence intervals for all policy-metric combinations
5	`descriptive_statistics.csv`	Descriptive statistics (mean, std, min, max, quartiles) for all experiments
6	`effect_sizes.csv`	Cohen’s d effect sizes for all pairwise policy comparisons
7	`exp1_summary.csv`	Experiment 1 summary: Policy comparison at K=20
8	`exp2_pivot_rt.csv`	Experiment 2 pivot: Mean response time by policy × fleet size
9	`exp3_pivot_rt.csv`	Experiment 3 pivot: Mean response time by policy × demand multiplier
10	`exp4_pivot_rt.csv`	Experiment 4 pivot: Mean response time by policy × service time mean
11	`optimization_comparison.csv`	Optimization model comparison (demand-weighted, p-median, maximal coverage)
12	`posthoc_comparisons.csv`	Tukey HSD post-hoc pairwise comparisons with corrected p-values
13	`queue_anova.csv`	ANOVA results for queue metrics across experiments
14	`queue_statistics.csv`	

#	Filename	Caption / Description
		Queue statistics (fraction queued, mean/max queue length) by experiment
15	`seasonal_analysis.csv`	Monthly seasonal variation analysis with factors and statistical tests
16	`sensitivity_summary.csv`	Overall sensitivity analysis summary across all experiments
17	`table1_baseline_comparison.csv`	Publication Table 1: Baseline policy comparison
18	`table1_baseline_comparison.tex`	Publication Table 1: LaTeX version
19	`table2_anova_summary.csv`	Publication Table 2: ANOVA summary
20	`table2_anova_summary.tex`	Publication Table 2: LaTeX version
21	`table3_pairwise_comparisons.csv`	Publication Table 3: Pairwise comparisons
22	`table3_pairwise_comparisons.tex`	Publication Table 3: LaTeX version
23	`table4_sensitivity_summary.csv`	Publication Table 4: Sensitivity analysis
24	`table4_sensitivity_summary.tex`	Publication Table 4: LaTeX version

Alternative Analysis Tables:

#	Filename	Description
25	`results/distance_comparison/comparison_table.csv`	Haversine vs Manhattan simulation comparison
26	`results/distance_comparison/allocation_comparison.csv`	Allocation differences by distance metric
27	`results/cbd_focused_comparison/comparison_table.csv`	CBD-focused vs Manhattan-wide performance
28	`results/cbd_focused_comparison/allocations.csv`	CBD-focused vs Manhattan-wide allocations

Capacity Sensitivity Tables:

#	Filename	Description
29	`results/capacity_comparison/full_comparison.csv`	Full capacity sensitivity comparison (all cap × K × policy)
30	`results/capacity_comparison/simulation_results.csv`	Simulation results for capacity experiments
31	`results/capacity_comparison/optimal_configurations.csv`	Optimal capacity configurations by policy
32	`results/capacity_comparison/allocation_statistics.csv`	Allocation statistics across capacity levels

Production V2 Tables:

#	Filename	Description
33	`results/production_v2/tables/descript-ive_statistics.csv`	V2 descriptive statistics (cap=2, P0-spatial)
34	`results/production_v2/tables/anova_results.csv`	V2 ANOVA results
35	`results/production_v2/tables/pos-thoc_comparisons.csv`	V2 pairwise comparisons
36	`results/production_v2/tables/confid-ence_intervals.csv`	V2 95% confidence intervals
37	`results/production_v2/tables/effect_sizes.csv`	V2 Cohen’s d effect sizes
38	`results/production_v2/tables/queue_statistics.csv`	V2 queue statistics
39	`results/production_v2/com-parison_with_v1.csv`	V1 vs V2 comparison table

Total: 39 table files (20 CSV + 4 LaTeX + 4 supplementary CSV + 4 capacity CSV + 7 V2 CSV).

---

12. Reproducibility

This section provides complete instructions for reproducing all results presented in this report.

12.1 Environment Specification

Component	Version / Value
Python	3.11.6
Operating System	Ubuntu Linux
NumPy	≥ 1.24.0
pandas	≥ 2.0.0
SimPy	≥ 4.0.0 (DES engine)
PuLP	≥ 2.7.0 (MIP solver, CBC backend)
SciPy	≥ 1.11.0
matplotlib	≥ 3.7.0
seaborn	≥ 0.12.0
geopandas	≥ 0.14.0
Shapely	≥ 2.0.0
folium	≥ 0.15.0
tqdm	≥ 4.65.0
openpyxl	≥ 3.1.0

Full dependency list: `requirements.txt`

12.2 Random Seeds and Reproducibility Strategy

All stochastic components use **deterministic seeding** for exact reproducibility:

Component	Seed / Strategy	Configuration
Base seed	`42`	`configs/demand.yaml → simulation.seed`, `configs/simulation.yaml → seed_base`
Production experiments	`SEED_BASE = 42`; replication i uses seed `42 + i`	`scripts/run_production_experiments.py` (line 52)
CBD experiments	`SEED_BASE = 42`; replication i uses seed `42 + i`	`scripts/run_cbd_experiment.py` (line 53)
Verification tests	Fixed seed `42` for all 4 tests	`scripts/run_verification.py`
Validation pilots	Pilot 1–2: seed base `100`; Pilot 3: seed base `200`; Pilot 4: `300 + rep`	`scripts/run_validation_pilots.py`
PRNG algorithm	NumPy PCG64 `Generator`	Used throughout via `np.random.default_rng(seed)`

Common Random Numbers (CRN): Dedicated random number streams are used for (1) NHPP arrival generation, (2) precinct assignment (multinomial), and (3) service-time sampling. Stream offsets ensure that changing one component (e.g., service time distribution) does not alter the arrival sequence, enabling valid variance reduction across policy comparisons.

12.3 Configuration Files

File	Purpose	Key Parameters
`configs/demand.yaml`	NHPP demand model parameters	Base rate (3.48/hr), lambda table paths, replications (30), seed (42)
`configs/service.yaml`	Travel time and service time models	Speed (20 mph), TOD factors, LogNormal(25, 10), dispatch delay (1.5 min)
`configs/optimization.yaml`	MIP optimization settings	Unit counts [20,30,40,48], capacity (5), threshold (8 min), CBC solver
`configs/simulation.yaml`	DES engine configuration	Horizon (168 hr), warmup (0), replications (30), seed base (42), threshold (8 min)
`configs/cbd_scenario.yaml`	CBD robustness experiment	Demand surge (2×), increased service times, CBD boundary, seed base (42)

12.4 Step-by-Step Reproduction Instructions

```
# 1. Clone repository and set up environment
git clone <repository-url>
cd ems-optimization
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# 2. Process raw data (requires raw data files in data/raw/)
python src/ems_readiness/data_processing.py
# Alternatively: make data

# 3. Run demand modeling
python scripts/demand_modeling.py

# 4. Run optimization comparison
python scripts/run_optimization_comparison.py

# 5. Run verification tests (4 tests)
python scripts/run_verification.py

# 6. Run validation pilots (3 pilots)
python scripts/run_validation_pilots.py

# 7. Execute production experiments (1,440 runs)
python scripts/run_production_experiments.py

# 8. Analyze production results
python scripts/analyze_production_results.py

# 9. Run CBD robustness experiment (330 additional runs)
python scripts/run_cbd_experiment.py

# 10. Run gap closure analyses
python scripts/analyze_queue_metrics.py
python scripts/analyze_seasonal_patterns.py

# 11. Generate publication figures
python scripts/generate_publication_figures.py
python scripts/generate_summary_dashboard.py

# 12. Run alternative analyses (distance metric comparison)
python scripts/generate_manhattan_distance_matrix.py
python scripts/run_distance_comparison_experiment.py --reps 10

# 13. Run alternative analyses (CBD-focused optimization)
python scripts/run_cbd_focused_optimization.py --reps 10
```

12.5 Data Requirements

File	Size	Source	Notes
`Motor_Vehicle_Collisions_-_Crashes_20260223.csv`	536 MB	NYC Open Data	2.24M records; not in Git (too large)
`FDNY_Firehouse_Listing_20260223.csv`	15 KB	NYC Open Data	219 firehouses; tracked in Git
`Police_Precincts_20260223.csv`	3.6 MB	NYC Open Data	78 precincts; not in Git
`manhattan_boundary.geojson`	5 KB	NYC Open Data	Tracked in Git
`cbd_boundary.geojson`	2 KB	MTA Congestion Relief Zone	Tracked in Git
`nyc_borough_boundaries.geojson`	650 KB	NYC Open Data	Tracked in Git

12.6 Expected Outputs

Successful reproduction generates:
- **44 figures** in `results/figures/`
- **24 table files** in `results/tables/`
- **Simulation logs** in `results/simulation/`
- All statistical results should match to within floating-point tolerance ($< 10^{-6}$ relative error) when using identical seeds and Python/NumPy versions.

End of Technical Report — Version 2.2.0 (Phase 10 — Enhanced Visualizations & CBD Coverage)